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Renewable Energy System — Complex Engineering Problem

ROOFTOP SOLAR POWER SYSTEM FOR SHAHZADI DAIRY FARM: A RENEWABLE ENERGY PROPOSAL

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Abstract

This report proposes an alternative energy solution for Shahzadi Dairy Farm (Katgora, Ashulia, Savar, Dhaka), which is currently energized by fossil-fuel-based power from the BREB line. A solar-only rooftop power system is designed around the farm's measured 24-hour load profile (8.8 units/day). The report covers the site and load demand, the choice of resource, the carbon footprint comparison, panel and battery sizing using worst-case (January) insolation, a cost estimate, and the societal and environmental impact of the transition.

Keywords: Solar PV; Dairy Farm; Load Profile; Carbon Footprint; Panel Sizing; Battery Bank; Savar, Bangladesh.

1 INTRODUCTION

The assignment requires proposing an alternative energy solution for a locality that is currently energized by fossil fuel-based power. For this, the chosen site is a dairy farm named Shahzadi Dairy Farm, located in Katgora, Ashulia, Savar, Dhaka. We attempted to present a renewable energy power system designed solely based on solar energy for this farm. The site is important and needs a switch to a renewable energy power system because lately the power consumption from the BREB line has increased much and the farm needs cost cutting measures.

1.1 Site Description

The shed area is 3200 sq.ft, which in meters is 297.28 sq.m of shed area with rooftop. Figure 1 shows the shed and its rooftop.



Figure 1: Shahzadi Dairy Farm: shed and rooftop area.

2 LOAD DEMAND ANALYSIS

To size the system, the farm's load demand was recorded over 24 hours by taking hourly readings from the energy meter (Figure 2). Table 1 lists the readings, and Figure 3 plots the cumulative load reading and the hourly usage derived from it.

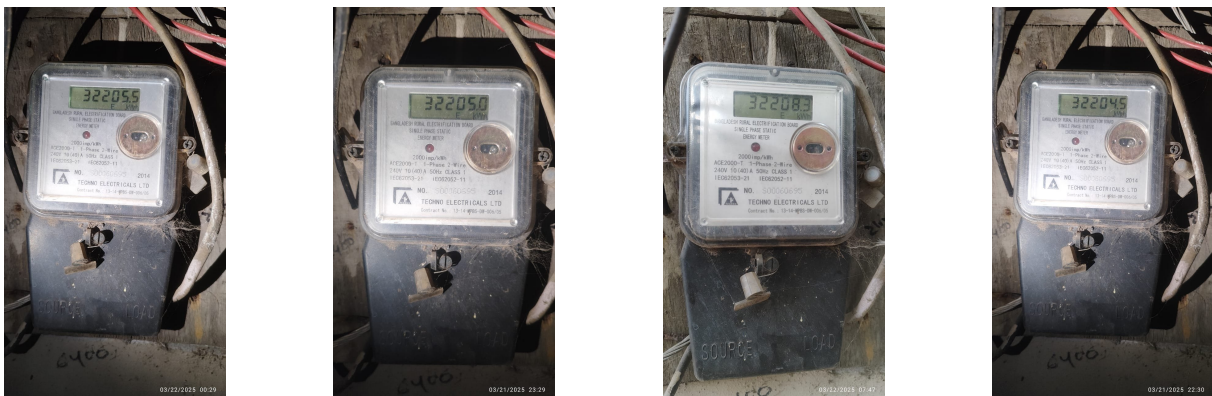
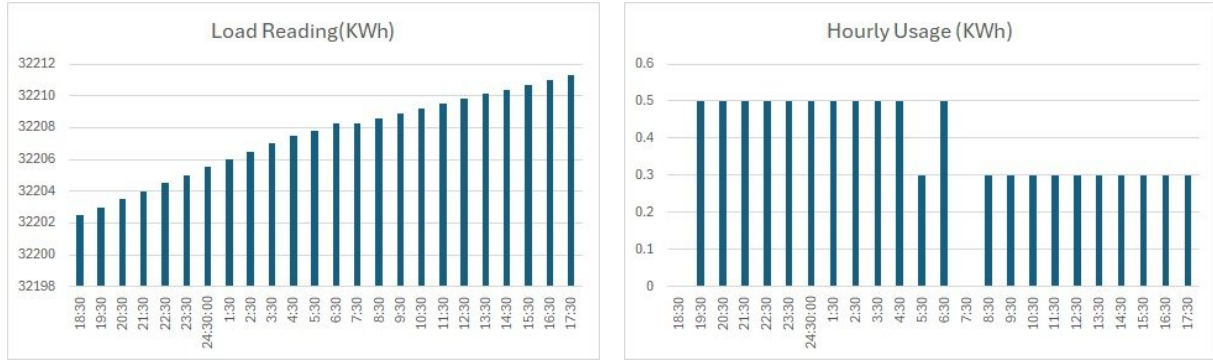


Figure 2: Energy meter photographs taken during the 24-hour load measurement.

Table 1: 24-hour load reading of the farm

Time (Hr)	Load Reading (kWh)	Time (Hr)	Load Reading (kWh)
18:30	32202.5	6:30	32208.3
19:30	32203	7:30	32208.3
20:30	32203.5	8:30	32208.6
21:30	32204	9:30	32208.9
22:30	32204.5	10:30	32209.2
23:30	32205	11:30	32209.5
0:30	32205.5	12:30	32209.8
1:30	32206	13:30	32210.1
2:30	32206.5	14:30	32210.4
3:30	32207	15:30	32210.7
4:30	32207.5	16:30	32211
5:30	32207.8	17:30	32211.3

**Figure 3:** Load reading (kWh) and hourly usage (kWh) over 24 hours.

The daily load demand is 8.8 units as from the 24 hour based load profile.

3 RESOURCE SELECTION

The type of cleaner energy resource available in that location is biomass from the farm. But we have not chosen that because the accommodation needed for the setup of a biomass based power plant is not enough and the owner of the farm does not want to give up any of his land for this purpose. So we have ultimately chosen to go with the solar energy based power plant.

4 CARBON FOOTPRINT ANALYSIS

4.1 Current Supply and Its Footprint

The farm is running on the power from the BREB line. So the fuel is fossil fuel.

The global average of the mix of oil, gas and coal based power generation is 0.5–0.7 kg CO₂ per unit of electricity production. We have taken the value to be 0.6 kg CO₂ per unit of electricity production. So, to produce 8.8 units of electricity the carbon footprint is

$$8.8 \times 0.6 = 5.28 \text{ kg CO}_2$$

4.2 Footprint with Solar Energy

If we use solar energy then the carbon footprint is on average 0.02–0.05 kg CO₂ per unit.

$$\text{Carbon footprint from solar energy} = 8.8 \times 0.04 = 0.352 \text{ kg CO}_2$$

Solar panels have **no operational emissions**—most emissions come from **manufacturing, installation, and recycling**.

4.3 Eliminating Life-Cycle Emissions

Proposed ways to eliminate this life-cycle emission:

1. Reducing the silicon waste, like monocrystalline panels are more efficient but energy-intensive; thin-film panels like CdTe have lower production emission.
2. Panel recycling programs.
3. Recycled aluminum frames.

5 PROPOSED ENERGY SOLUTION

We have proposed a solar energy based power system on the rooftop of the farm. Figure 4 shows the system architecture: the PV cells feed the inverter (which supplies the AC loads) and the charge controller, which charges the battery bank.

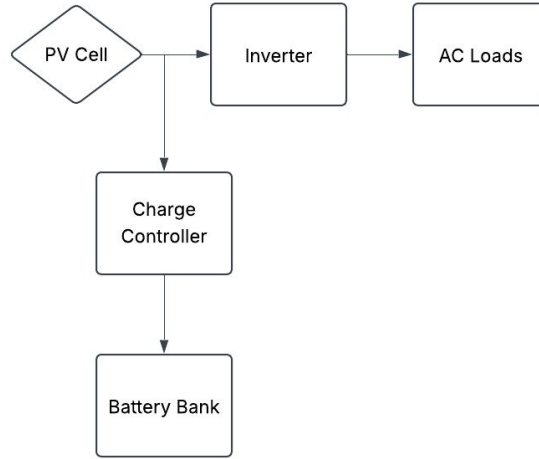


Figure 4: Block diagram of the proposed solar power system.

5.1 Solar Irradiance Data

Figure 5 shows the monthly total irradiance for the site. Source: (<https://www.nshd.sreda.gov.bd/solar-irradiance-data>).

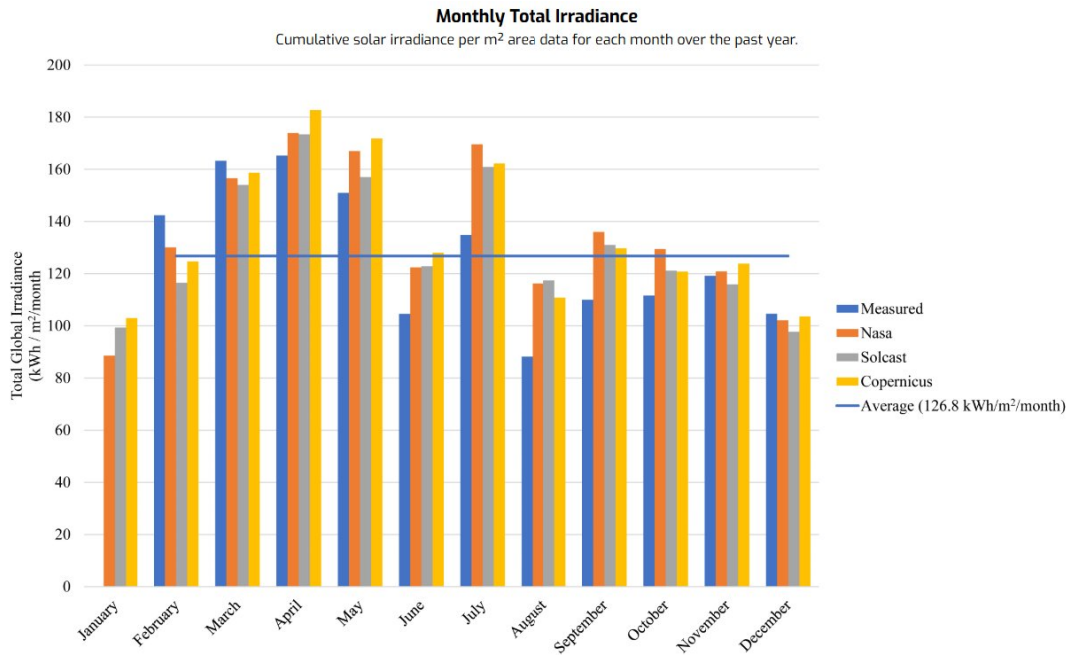


Figure 5: Monthly total irradiance (SREDA).

5.2 Chosen Solar Panel

The chosen solar panel is the **Longi 580W N-Type Solar Panel**, product code LR5-72HTH-580W (<https://ipsbazarbd.com/product/longi-580w-n-type-solar-panel/>). Its spec-

ification is shown in Figure 6.

Identification	
Brand	LONGI
Product Code	LR5-72HTH-580W
Design & Mechanical Specification	
Panel Type	Bi-facial
Panel Surface	Standard
Panel Weight	27.5 Kg
Panel Length	2278.0 mm
Panel Width	1134.0 mm
Panel Thickness	35.0 mm
Cell Material	Monocrystalline
Cell Type	Half-cut Cell
No. of Cells	144.0
Cell Dimension	N/A
Glass Type	Single

Figure 6: Longi LR5-72HTH-580W datasheet.

6 SYSTEM SIZING

6.1 Solar Panel Calculation

From the data table, daily load demand is 8.5 kWh.

From the solar irradiance data from SREDA, the average insolation is 128 kWh/m²/month. But to make our calculation efficient, we have taken the worst case scenario into account. The worst insolation is for the month of January, which is 84 kWh/m²/month. So for a day we have

$$\frac{84}{31} = 2.709 \text{ kWh/m}^2/\text{day}$$

So, 2.709 kWh is obtained from 1 sq.m; 8.8 kWh is obtained from

$$\frac{8.8}{2.709} = 3.248 \text{ sq.m}$$

From the panel data sheet: panel length = 2278 mm, panel width = 1134 mm, panel area = 2.58 m².

Now, if efficiency is 100% then the area is 3.298 sq.m; if efficiency is 22.5% then the area is 14.43 sq.m. Hence

$$\text{Number of panels} = \frac{14.43}{2.58} = 5.59 \approx 6 \text{ panels}$$

6.2 Battery Calculation

$$Wh_{load(battery)} = \frac{Wh_{night}}{\text{efficiency}} = \frac{6.1}{0.7} = 8.7143 \text{ kWh} = 8714.3 \text{ Wh}$$

where the efficiency of the battery is 70%.

$$Ah_{bat} = \frac{Wh_{load}}{DOD \cdot V_{bat,nom}} = \frac{8714.3}{0.7 \times 24} = 518.71 \text{ Ah}$$

where DOD = 0.7 and $V_{bat,nom} = 24 \text{ V}$.

7 COST AND ECONOMIC ANALYSIS

The cost for the solar panel is 10730 Taka per panel. For 6 panels we will need $6 \times 10730 = 64380$ Taka. The inverter cost will be around 50K–60K Taka and the cost of the MPPT device will be around 30–35K Taka. The battery cost will be minimum 1 to 1.5 lakhs Taka depending on the materials used.

Table 2: Estimated minimum setup cost

Component	Cost (Taka)
Solar panels (6×10730)	64,380
Inverter	50,000
MPPT device	30,000
Battery	150,000
Total minimum cost	294,380

So the total minimum cost for setup is $64380 + 50000 + 30000 + 150000 = 294380$ Taka. Rounding the figure, the setup cost will be 3 lakhs Taka including other costs like wiring and hardware cost.

The monthly bill of the BREB line is 6K Taka. Though the set up cost is higher, we will be able to cut cost by minimum 60–65K Taka per year. The panel performance warranty is 25 years. So in the long run the owner will be benefitted.

8 IMPACT ON SOCIETY AND ENVIRONMENT

The impact of this energy transition is huge given that the owner can cut cost in his business and save money, and the environment will be less polluted than before. If all over the country

this can be done then the dependence on fossil fuel will be reduced a lot. The country can save foreign reserves, boosting the economy. Global warming will be reduced.

9 CONCLUSION

From this assignment we got to know how a farm needs power all over the day and how the resources can be utilized. We saw that though we had biomass, we could not use that, which gave us an overview of the practical scenario. Furthermore, we got to practically apply our acquired knowledge all over the course.